



SupremeRAID™ by GPGPU

Product Introduction

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SupremeRAID™

4th Revolution of GPGPU Application
Highest NVMe Data IO for Big Data protection



The New Standard in Enterprise Data Protection & Performance

- Architect Innovation “ No Cabling Raid Card” and rely on GPU computing power
- The Most Safety Data protection
- Highest Performance NVMe SSD RAID
- Most flexible: No additional cable, no BBU, support 32 NVMe SSD

HQ in Silicon Valley

R&D center in Taipei, Taiwan

Global Network of Partners,
OEMs, Distributors & Resellers



The Safest SSD RAID Card

1. **Support RAID Levels:** 0/ 1/ 5/ 6/ 10
2. **Controller Failover HA**
3. **Double Failure Protection with Distributed Journaling:**
2nd Journal in SSDs when RAID crush.
4. **Coming soon in 2025:**
Non-disruptive rebuild (Data backed up to a new SSD before failure), and erasure coding

Brilliant Figure with SupremeRAID™

Since 2021 Graid named
over 10 famous awards in worldwide



2023 APICTA

APICTA Named **Golden Award** in BS category. Judges were especially impressed with its ability to maximize NVMe SSD performance in AI and high-performance 化 NVMe SSD

- Hong Kong



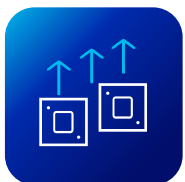
2024 Flash Memory Summit

Joint solutions with partners won **the Most Innovative Enterprise Business Application** and **Best of Show awards** for excellence in high-performance media and entertainment applications.

- The U.S.

Irreplaceable SupremeRAID™ for efficient AI/HPC/database applications

The advantages and features make it the only best choice for customers!



Liberate CPU resource

Release CPU resources to
process high workload



World Record Performance

4K Random Read **28M IOPS**
1M Sequential Read **260 GB/s**



Most Safety data

RAID 0/1/5/6/10
Controller Failover HA
Double Failure Protection
with Distributed Journaling



Best Raid solution for VM

**3~5X performance boost in
Random IO**
**15 x Bandwidth increase in
Write**



AI& HPC Solution

SupremeRAID™ + NVMe SSD
**2~5 times performance
improvement**
(Lustre, BeeGFS)

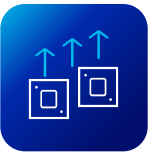


Perfect solution for Database

More than **2x performance**
improvement
saving 7x rebuilding time
Increase **150% capacity** in RAID6

SupremeRAID™ By Graid Technology

Battle Card



Liberate CPU resource
Release CPU resources to process high workload



Most Safety data
Raid 0/1/5/6/10
Controller Failover HA
Write-hole Protection



Highly Scalable
Supports **NVMe-oF**
up to **32** NVMe drives supported



Best Performance
4K Random Read **28M IOPS**
1M Sequential Read **260 GB/s**

Replace Traditional Raid

- ✓ Double performance for 4 Gen5 NVMe SSDs compare with Broadcom 9670.
- ✓ Supports up to 32 NVMe drives in RAID with 28M IOPS
- ✓ **SupremeRAID™ supports two RAID cards in HA mode**

Best Raid Solution for VM

- ✓ **Boost Random Read/ Write performance 3~5X**
- ✓ Increase Write bandwidth more than 15 times
- ✓ **Stable performance** as VMs grow
- ✓ **Reduces CPU workload** significantly

Best performance of File system in AI / HPC

- ✓ Lustre+Graid write bandwidth is over 2 times higher than traditional ZFS
- ✓ BeeGFS+ Graid write bandwidth is over 5 times higher than traditional hardware RAID cards. **OSS nodes can be halved**
- ✓ Reduce customer TCO

Perfect solution for Database

- ✓ More than **2-5x** performance improvement
- ✓ saving **7x rebuilding time**
- ✓ Increase **150% capacity** in RAID6

Usecase/ Applications (Target to HPC applications customer)

Lustre Cluster



synchrotron radiation

BeeGFS Cluster



MGI

3D modeling



NLSI

Deep Learning



ERP System



ZOHO

Smart meter



Taiwan Power Company

E-commerce



Toyo Bank

Cyber security



CHT

Broadcasting CG creation



N H K

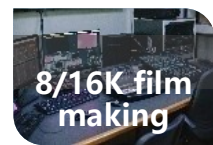
Virtualized cloud database



NIPA Cloud



NASA



8/16K film making



CAD

Potential Market and Eco-system



PROTECTING YOUR DATA FROM **THE EDGE TO THE CLOUD**



Enhanced Data Protection



Unbeatable Performance

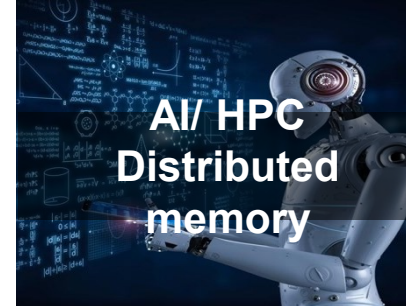


User Friendly



Not only a RAID card But a comprehensive big data eco-system

Cyber Security for Big Data



Big Data Transfer

High Performance Storage Pool

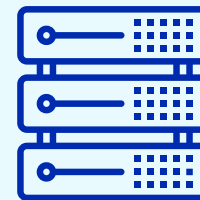
Throughput:
260GB/S
High IOPS:28M

lustre®



BeeGFS

SDS/ File System



Server



High Performance Data Protection



Graid
SupremeRAID™



NVMe SSD

Use case

Database

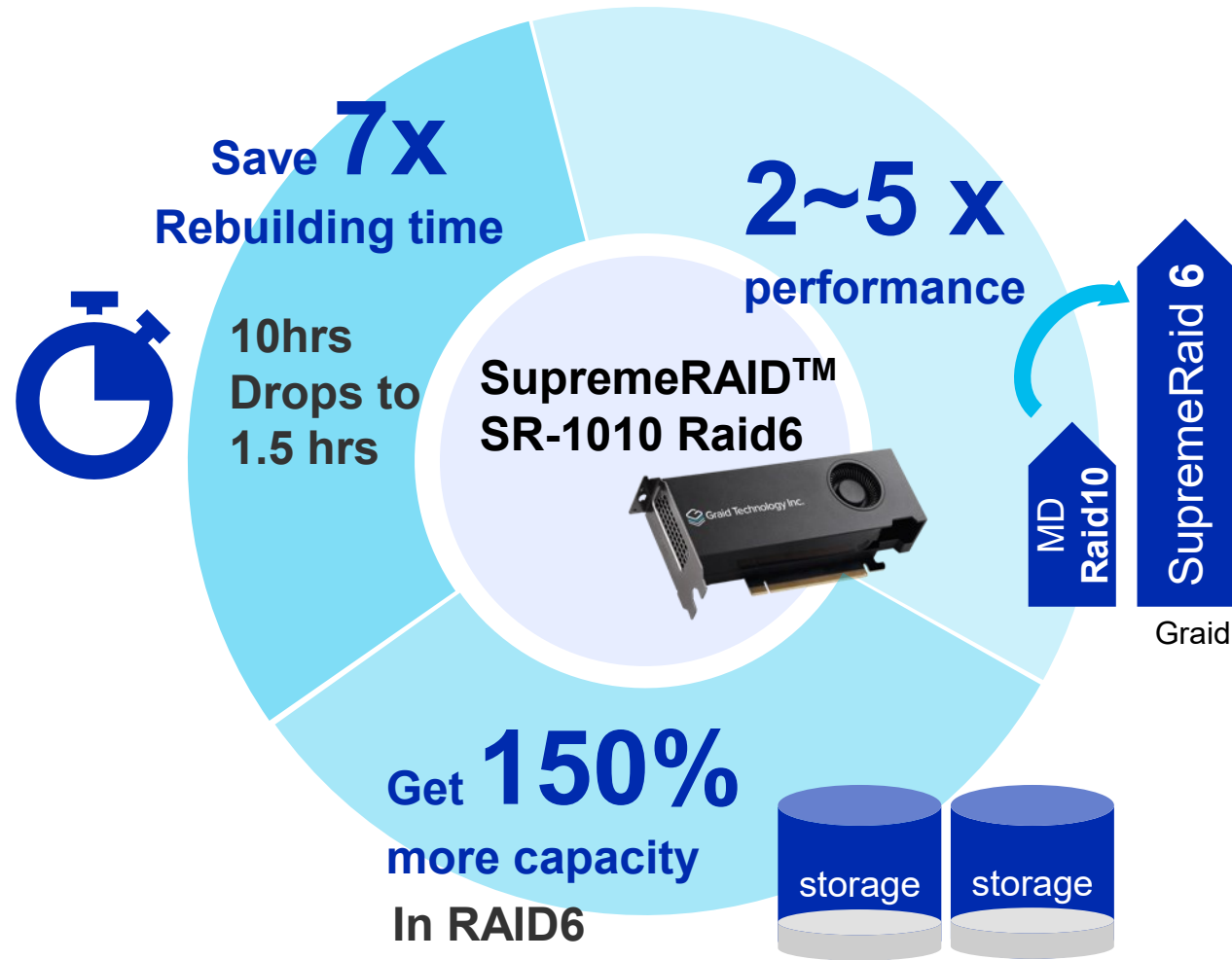
Local Databases

“Current hardware RAID is obviously not the solution for local databases due to its latency and performance limitation for NVMe SSDs. SupremeRAID™ provides **unbeatable performance** and **latency** for databases to realize the full benefits of using NVMe SSDs.”

— Solution Architect, Tier-one IT Company



Advantages with using SupremeRAID™



Use case

Database in E-commerce



E-Commerce



ERP system

Web portal/ purchasing/ manufacturing/
time& project/ distribution



Payment System

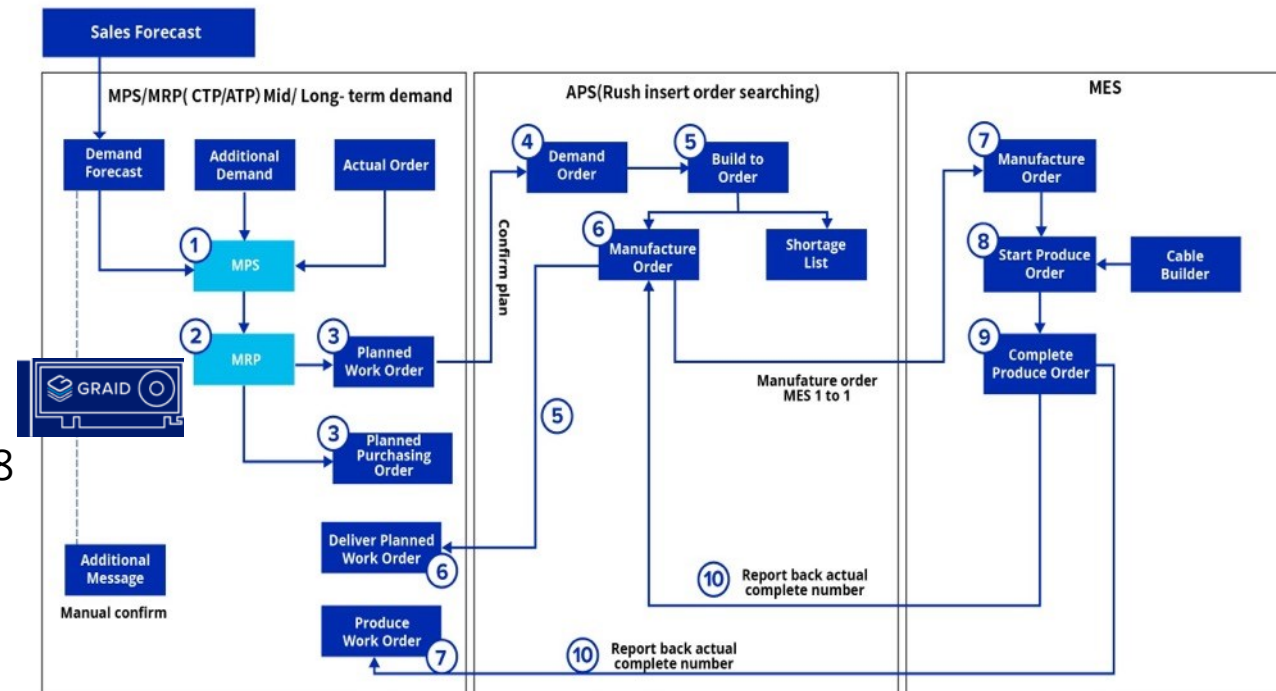
Monitoring/ capture/ storage after
decoding



Cyber security

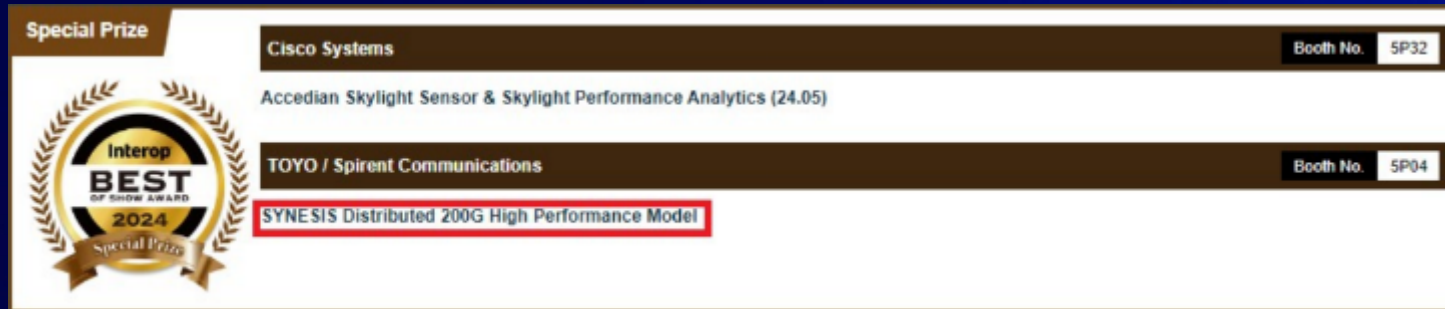
Smart Factory: ERP system with Oracle Database

1. **ERP:** Enterprise Resource Planning / **APS:** Advanced Planning and Scheduling / **MES:** Manufacturing Execution Systems
2. **MPS:** Master Production Schedule/ **MRP:** Material Requirement Planning
3. More than **100M** data R/W analysis in MPS and MRP
4. Oracle database and Hitachi/ EMC storage used to use
5. IOPS was too slow and high latency caused more than 8 hrs every week for production plan.
6. With Graid SR-1001-ED4 + 4 PCS Phison SSD, **shorter the production plan to 5 hours**



Payment System in E-commerce

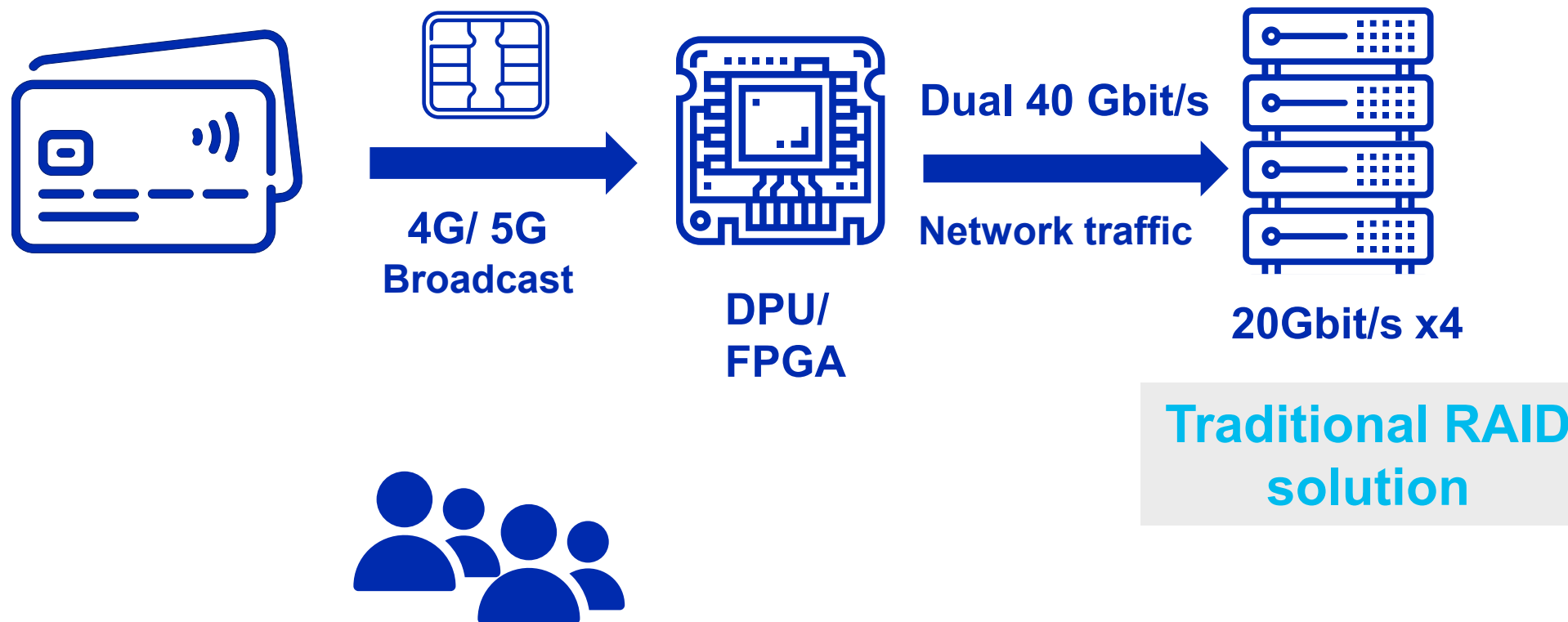
E- Commerce : Packet Capturing Payment System(Toyo)



Best of Show Award at INTEROP Tokyo 2024
Interop Tokyo is an Internet technology event

Finance- Payment System

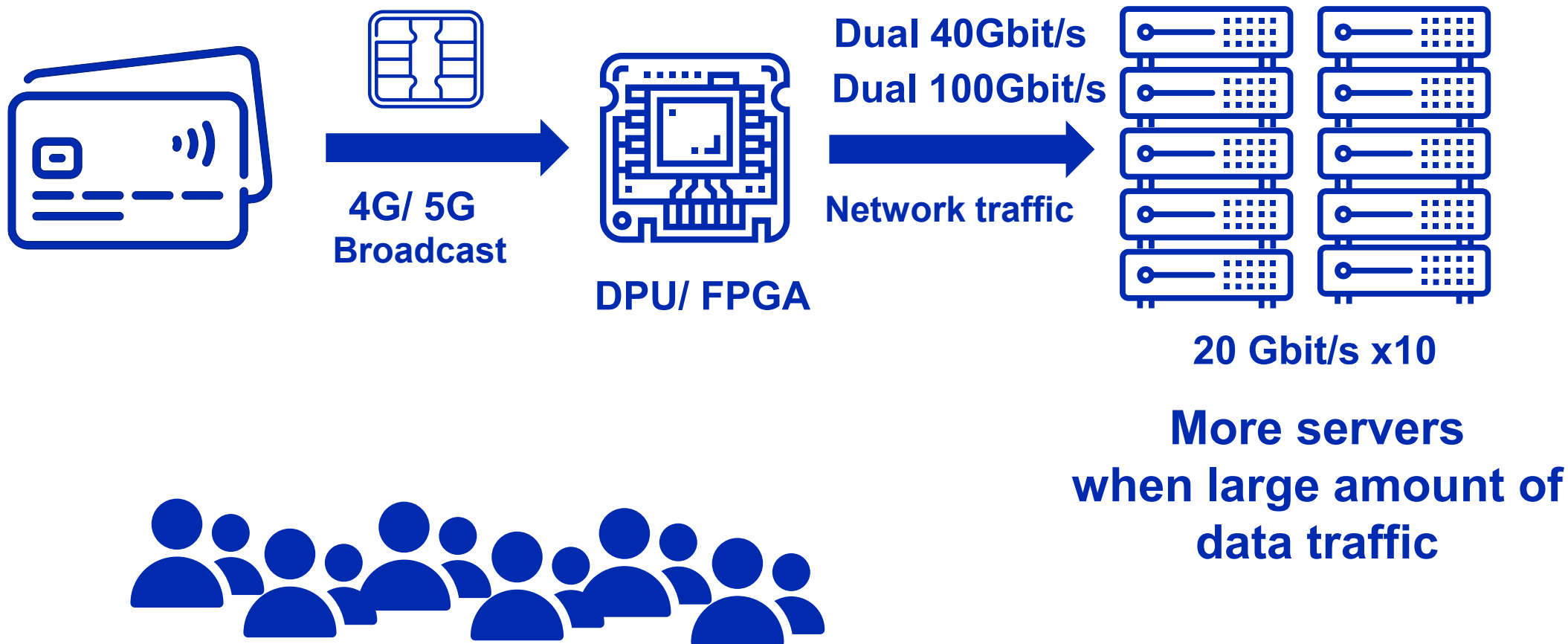
BEFORE



Finance- Payment System

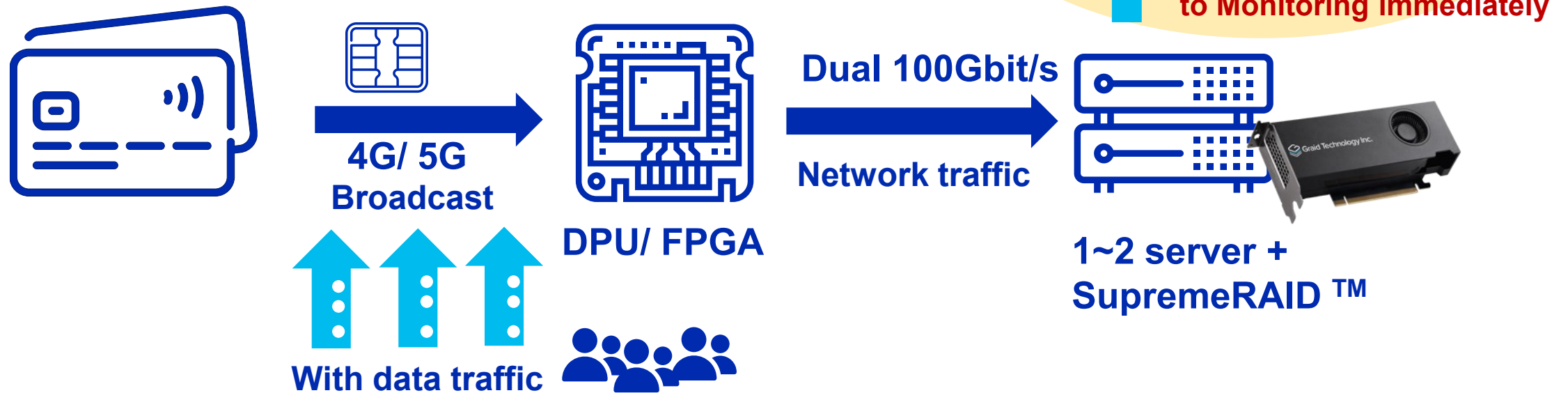
NOW

Purchase behavior with e-commerce Increased



Finance- Payment System

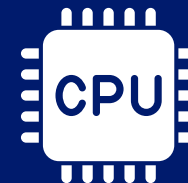
NOW + SupremeRAID™



Capture



Storage



Monitoring



Cyber Security in E-commerce

Cyber security: SIEM in Elastic Database



CHT SIEM

Project requirement: Provide real-time analysis of security alerts generated by applications and network hardware. They are also used to log security data in **500K eps** and generate reports for compliance purposes.

Pain point:
SIEM appliance can only offer 50K eps. Dozens of all-flash memory servers are needed to barely meet storage performance.

Graid Solution:
SR-1010-FD32 all Flash server can receive over **200K EPS in**

Elastic DB

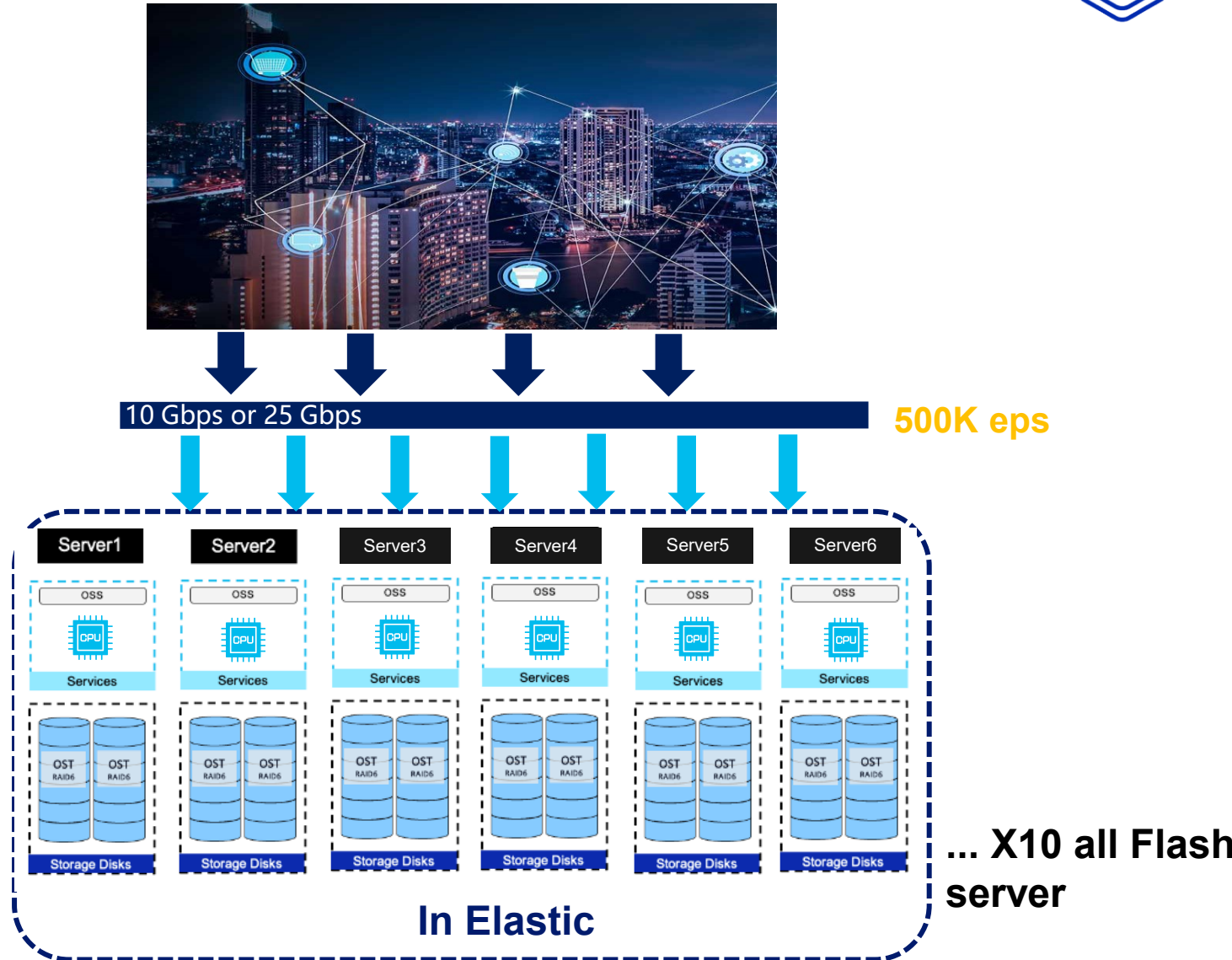
Benefit:

- 5 times performance upgrade in Elastic
- Less Space
- Less Power
- Lower TCO

Traditional solution

Less than 50K eps
per node
Hard to meet

Using over 10
servers to meet



Cyber security: SIEM in Elastic Database



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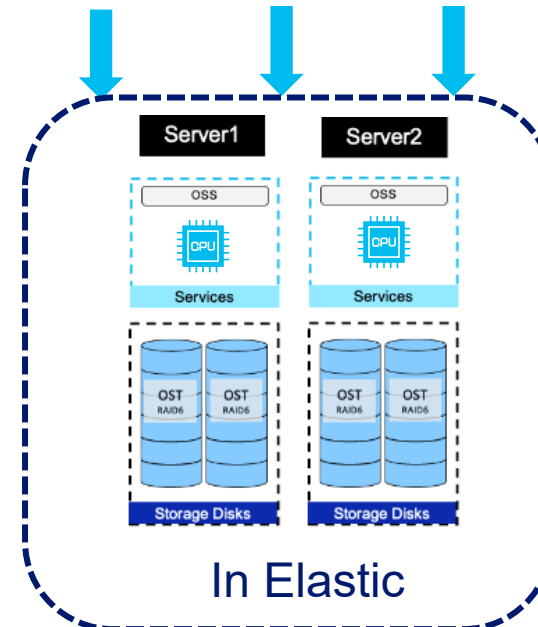
Graid Solution

Up to 250K eps per node
Easy to meet **500K eps in 2 nodes**



10 Gbps or 25 Gbps

500K eps

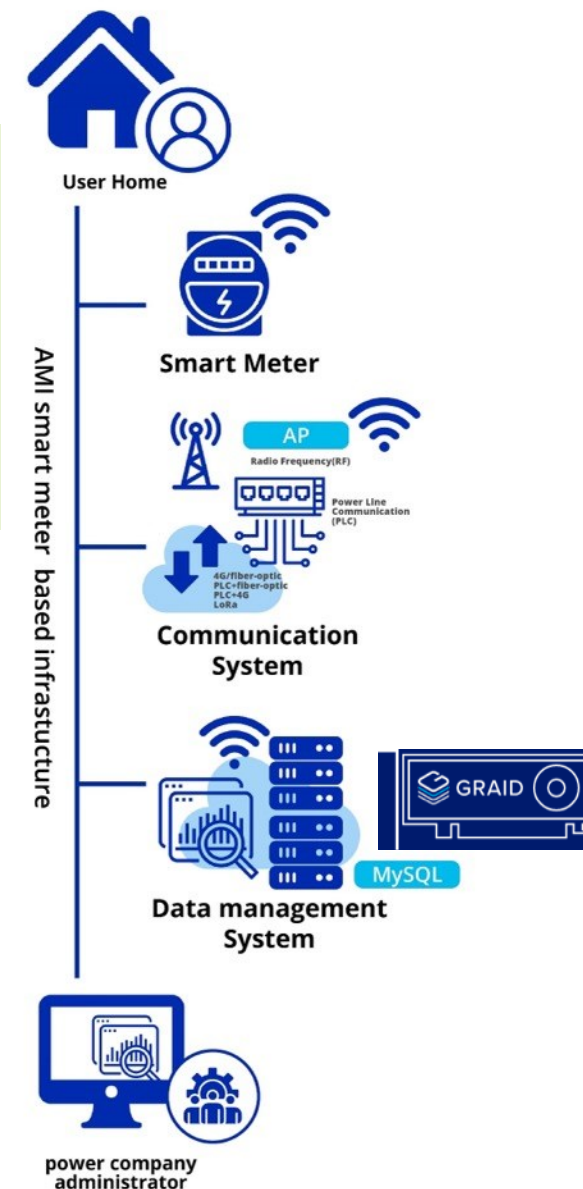
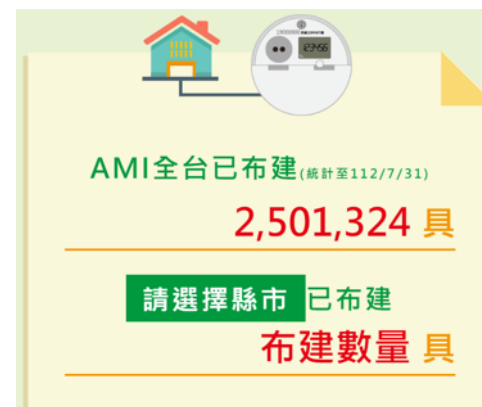


Benifit:

- ✓ Less Power
- ✓ Less Space
- ✓ Lower cost

Advanced Metering Infrastructure,

1. Smart meter is a modern power measurement device with **communication function**.
2. Smart meters are the core technology for the development of "smart grids" around the world, and the large amount of data generated by smart meters can effectively **improve the reliability** of power supply, **improve power transmission efficiency**.
3. smart meter replacement operations in 2020, till now completed more than **2.5 million units** and are moving towards the goal of completing **3 million units** in the year 2024.
4. Performance with Software RAID can't meet requirement.
5. **Two times improved with SupremeRAID™ solution.**



Aerospace database(NASA)

NASA maintains databases across **more than 7 distinct domains**, with a continuous influx and outflow of vast amounts of data daily.

Simultaneously, there is a stringent requirement for extremely high levels of security and rapid read performance. The traditional RAID cannot meet the requirement.

Graid Solution: :

- 1.saving **7x** rebuilding time
- 2.More than **2x performance** improvement
3. **150% capacity** increased



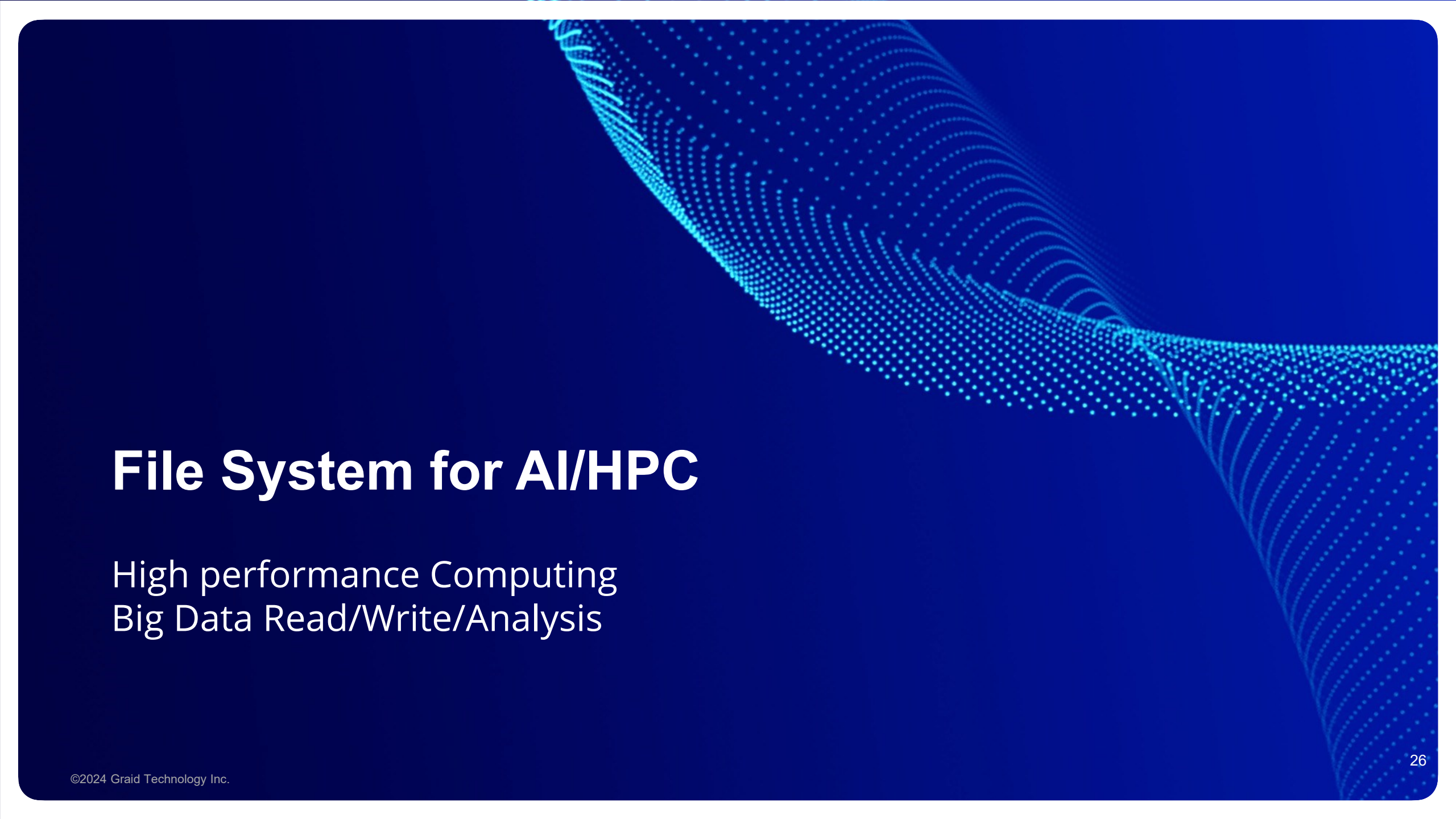
Cloud DB service

- Low Latency and higher performance are most important for databases, especially for hosting companies.
- Can't meet customer's requirements Even with RAID 10 implemented.
- 50% usable capacity in RAID 10, hard to sustain for hosting business.

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File System for AI/HPC

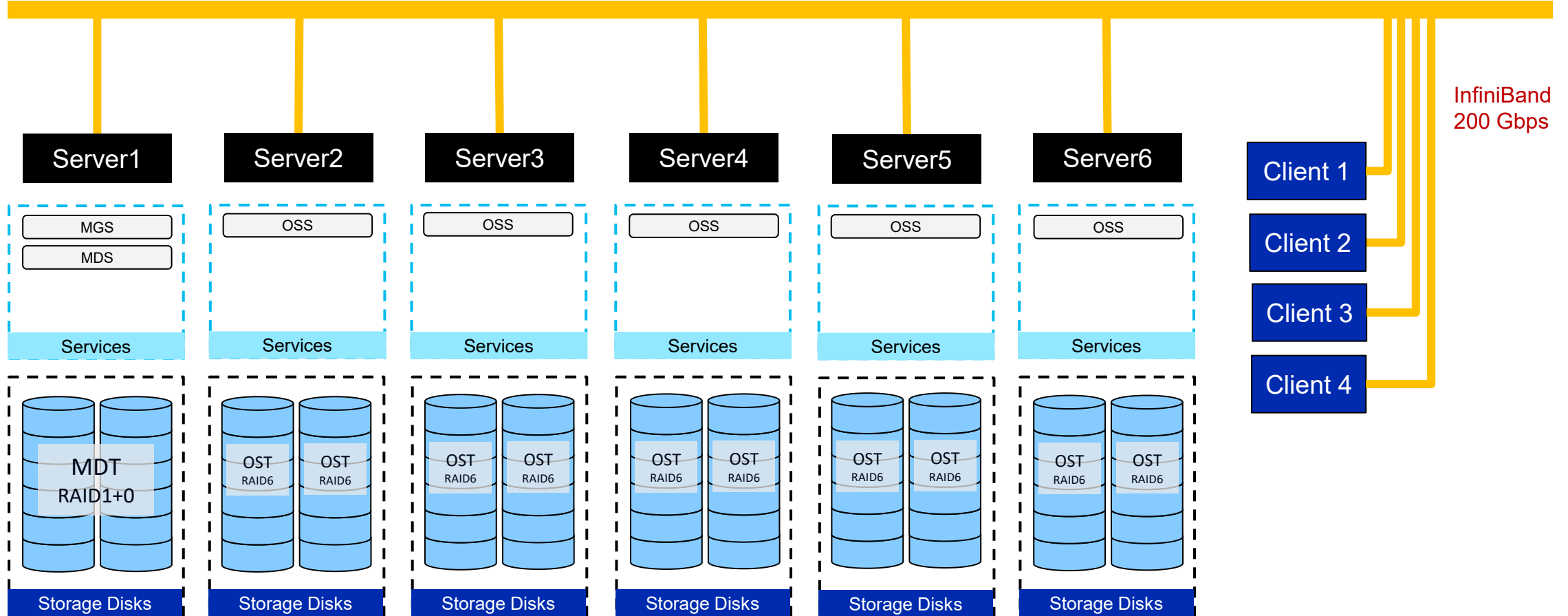
High performance Computing
Big Data Read/Write/Analysis

File system for AI/HPC

HPC Cluster

- a. Lustre
- b. BeeGFS

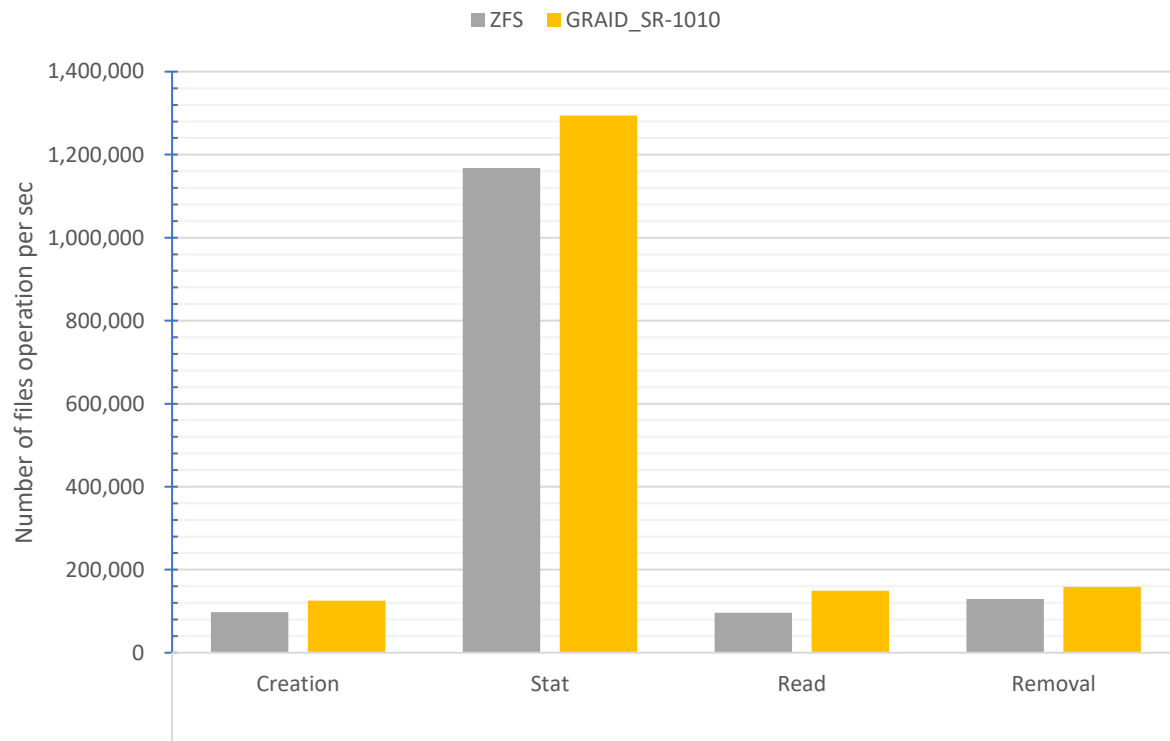
Architecture of Lustre



- **MDT (Metadata Target)**
We created two disks to form a single mirror group and combined six mirror groups into one RAID0 group.
- **OST (Object Storage Target)**
We created a RAID6 group with six disks on the same NUMA node, with two OSTs per OSS (Object Storage Server).

Lustre Performance

Lustre MDTEST for file Creation, Stat, Read and Removal



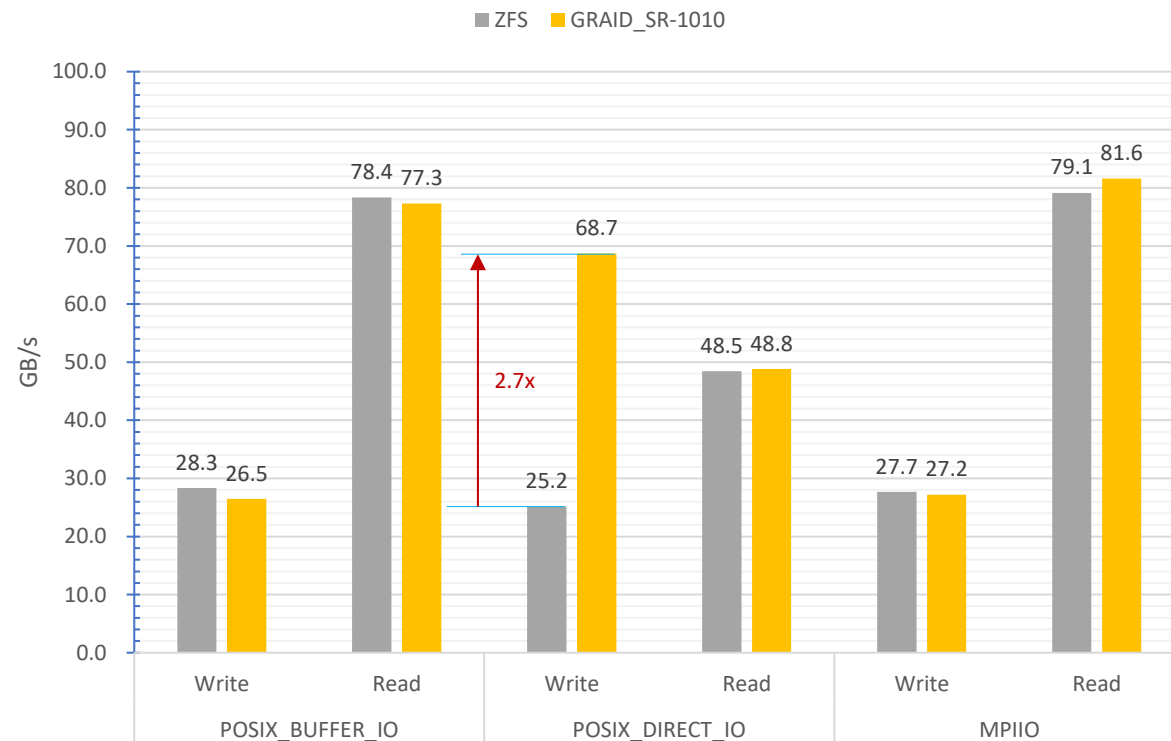
- **MDTEST**

MDTEST performance test results indicate that when using SupermeRAID SR-1010 as the RAID controller, the metadata performance is only slightly better compared to ZFS, with no significant overall performance difference.

- **IOR**

IOR performance test results show that there is no significant difference in Buffer I/O performance. However, when using \SupermeRAID SR-1010 as the RAID controller, the write performance of Direct I/O has an advantage of **more than 2.5 times compared to ZFS**.

Lustre Max Throughput



Synchrotron Radiation Research

Pain point:

Build a storage system with **≥60GB/s** write performance in minimal space to protect data and ensuring success of research.

Solution:

4 x 2U servers with 12 NVMe SSDs each, using Lustre with Graid for RAID protection, aggregated performance more than 60GB/s.

Features:

Bandwidth Performance:

Single node achieves 15GB/s write bandwidth, over 100% improvement compared to software or traditional RAID; Lustre aggregates to 60GB/s.

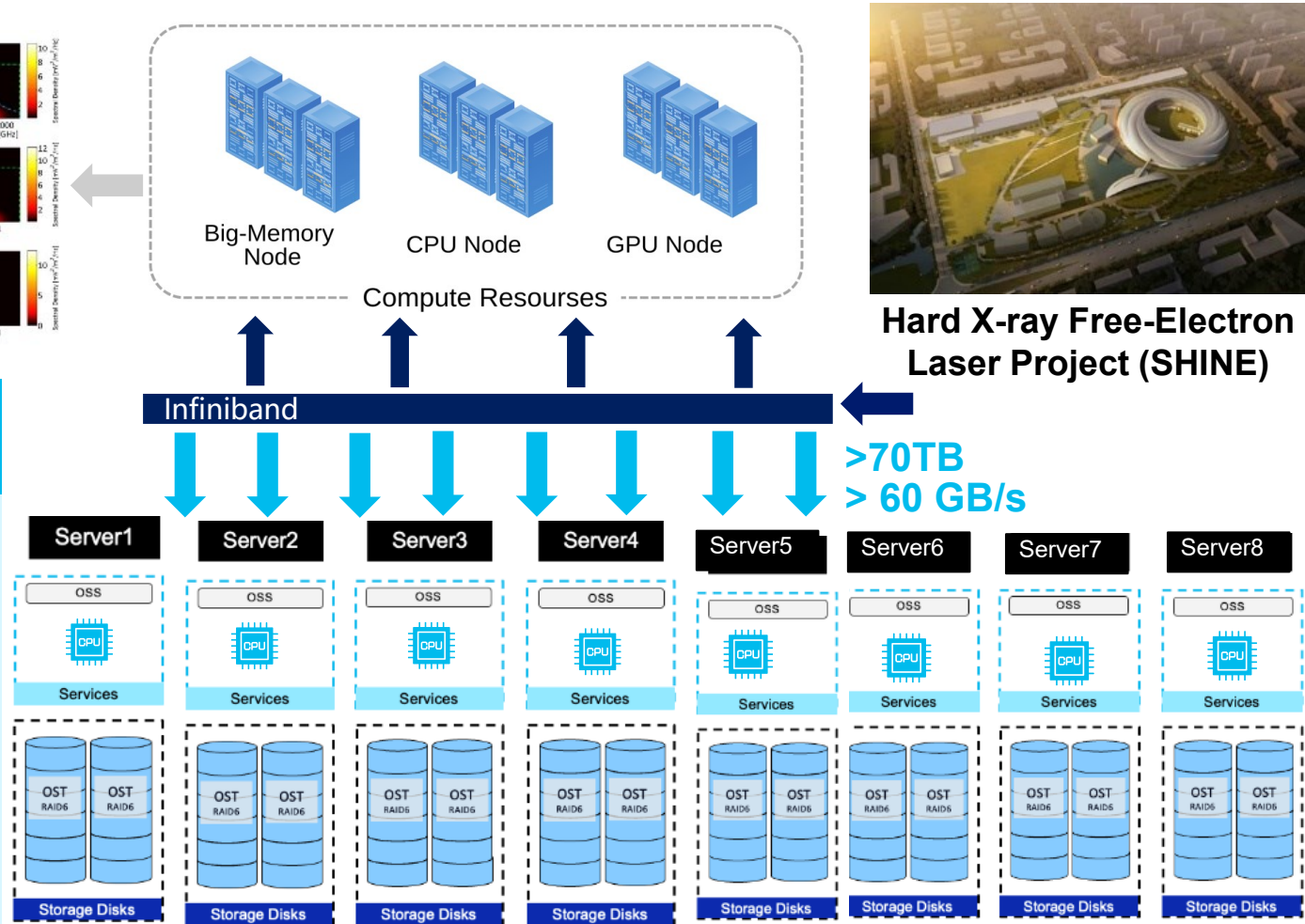
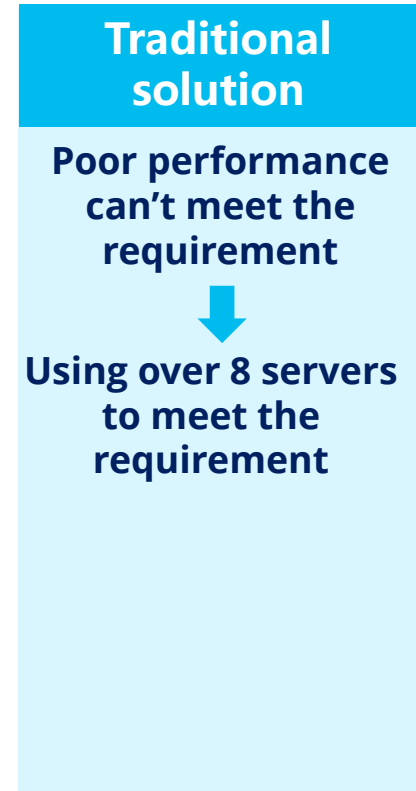
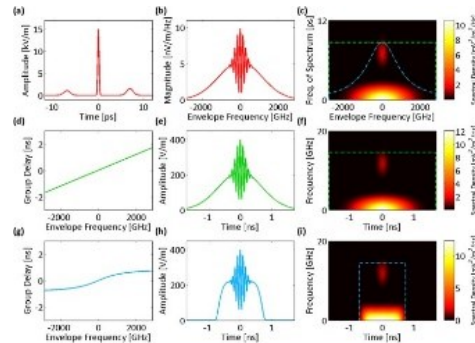
Cost Savings:

Over 50% savings, reducing from 8+ devices to 4.

Space Savings:

Reduces from 16U to 8U.

Lustre + SupremeRAID™: Over 2X write performance compared with ZFS



occupy space and high-power consumption

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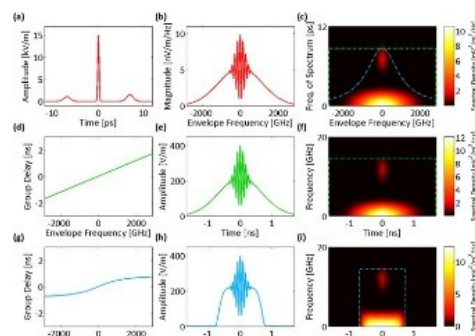
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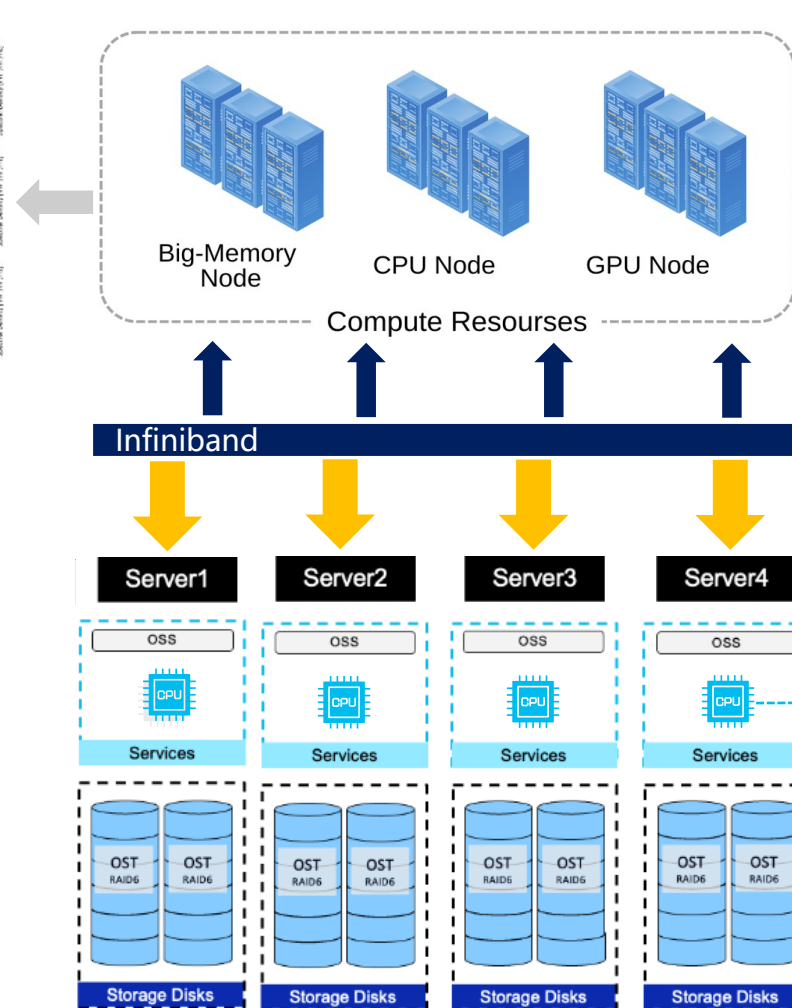
Traditional solution

Poor performance can't meet the requirement

Using over 8 servers to meet the requirement

With SupremeRAID™

Can deliver **60 GB/s** performance with **4 servers**



Hard X-ray Free-Electron Laser Project (SHINE)

>70TB
> 60 GB/s

SupremeRAID™ unleashes the CPU resource and delivers 100% SSD performance

Benefits:

- ✓ Electricity Saving
- ✓ Space Saving
- ✓ Cost lowering



SupremeRAID™+ Lustre

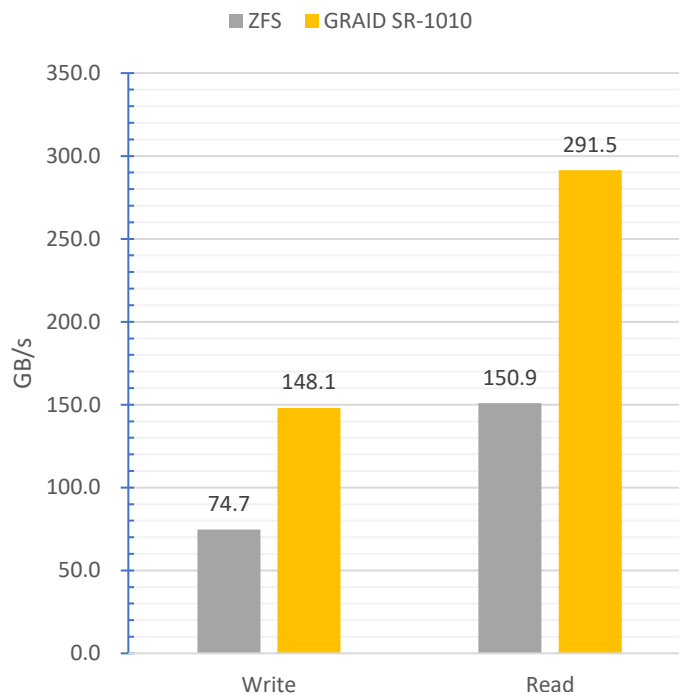
File system for AI/HPC

HPC Cluster

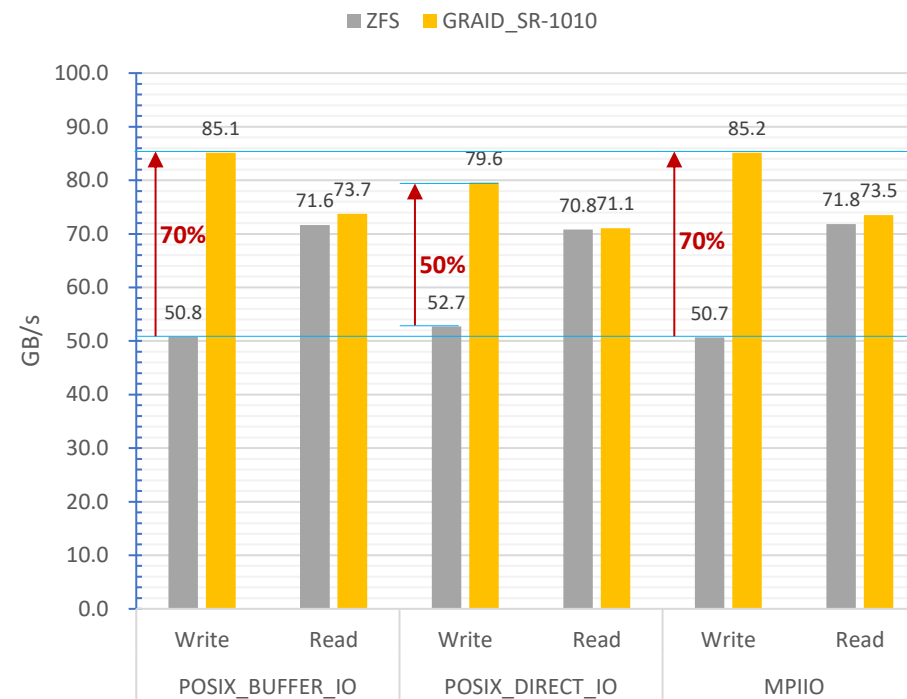
- a. Lustre
- b. BeeGFS

BeeGFS Performance → 2~2.5x Lustre Performance

BeeGFS StorageBench



BeeGFS Max Throughput



- **BeeGFS StorageBench**

Compared to ZFS, when using SupremeRAID SR-1010 as the RAID controller, **the overall performance of OSTs increased by approximately 2 times.**

- **MDTEST**

In terms of Metadata performance, when using SupremeRAID SR-1010 as the RAID controller, the performance of File **Creation, Read, and Remove operations improved by more than 2 times.**

- **IOR**

Regarding read and write performance, when SupremeRAID SR-1010 is used as the RAID controller, read performance is similar to ZFS, but write performance has seen an average increase of 50 to 70%, approaching the bandwidth limits of InfiniBand HDR networks (100GB/s).

Deep Learning

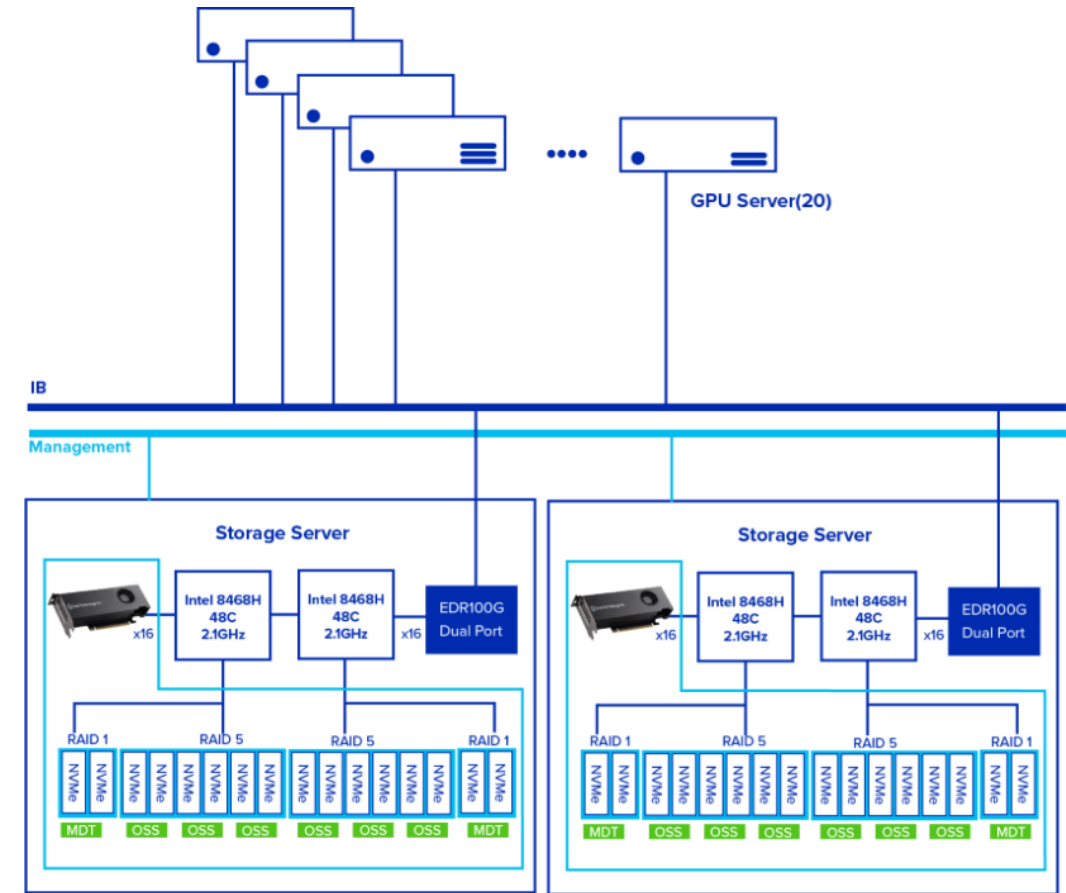
SupremeRAID™ improves deep learning training efficiency

Challenge: The current storage bottleneck has resulted in high latency

Solution: Using multiple GPU servers for deep learning, a fully flash memory server + SupremeRAID™ provides high-performance, low-latency data transfer

Result:

- Compared to traditional software RAID, performance has increased by **11 times** (from 8 Gbps to 88 Gbps)
- Training time has been **reduced by 20%** (from 10 hours to 8 hours)



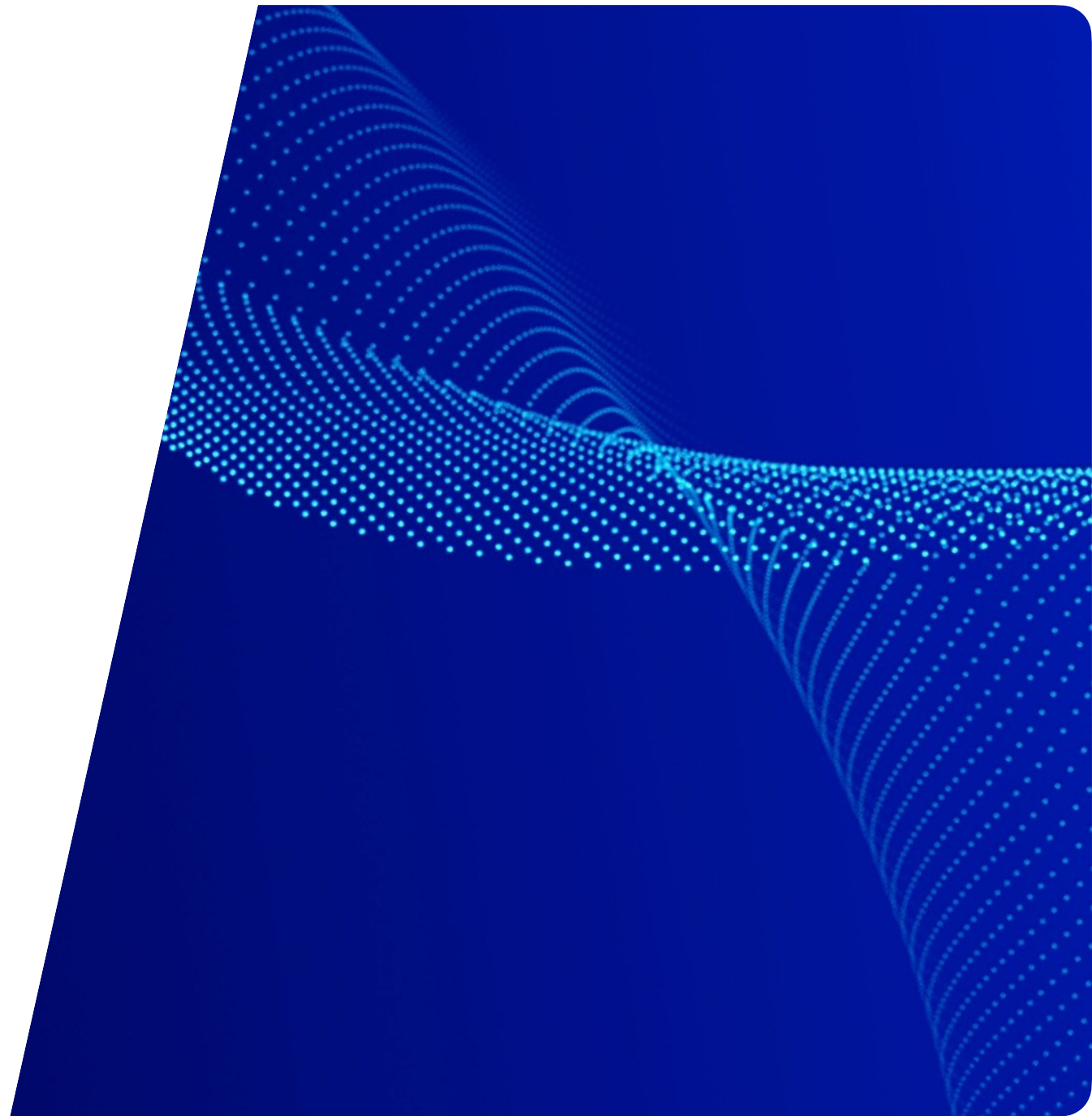
Test Environment Specification:

- Rocky Linux 8.7
- IB EDR 100Gb NIC x 2
- NFS server

USE CASE: AI /HPC

**Edge computing with
Workstation**

Data IO Bottleneck



SupremeRAID™

Solving Data IO bottleneck and saving over 2x processing time

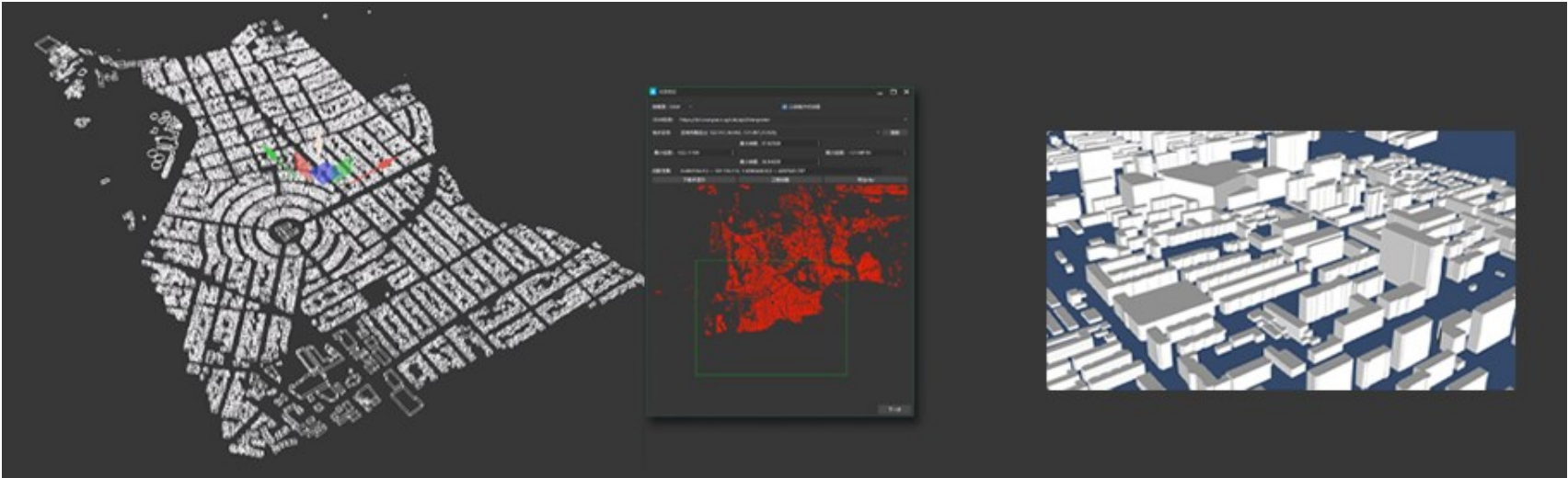
3D Modeling

Background: Map service conducting 3D geographical modeling, using the workstation Xeon + SAS HDD RAID and SVSGeoModeler software

Pain point: Poor performance results in a long processing time. Even increasing the CPU cannot improve overall performance

Solution: Use Intel i9 CPU +SupremeRAID™ Solution with 4 NVMe SSD to enhance 2 times performance.

- Features:**
- Improve the data IO performance will upgrade computing performance and save the half computing time.
 - After breaking data IO bottleneck, upgrade CPU performance can enhance overall performance more.



	Workstation Xeon + SAS HDD RAID	Intel i9 CPU + SupremeRAID™ + 4NVMe SSD	
Free Mesh Space	3hrs 3mins	1hr 17mins	2.3x time saving
Constrained Mesh Space	1hr 16 mins	32mins	2.3x time saving
3D Model Creation	108hrs12 mins	55hrs 30mins	1.9x time saving

Perform CAD operations with Ansys

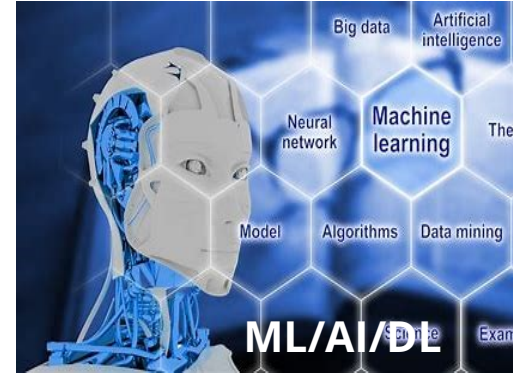
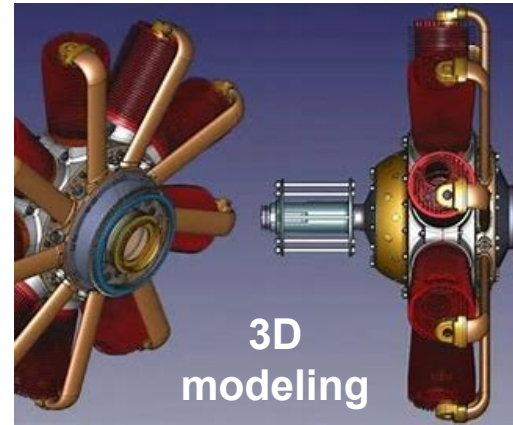
SupremeRAID™ reduces computing cost for computer-aided design (CAD)

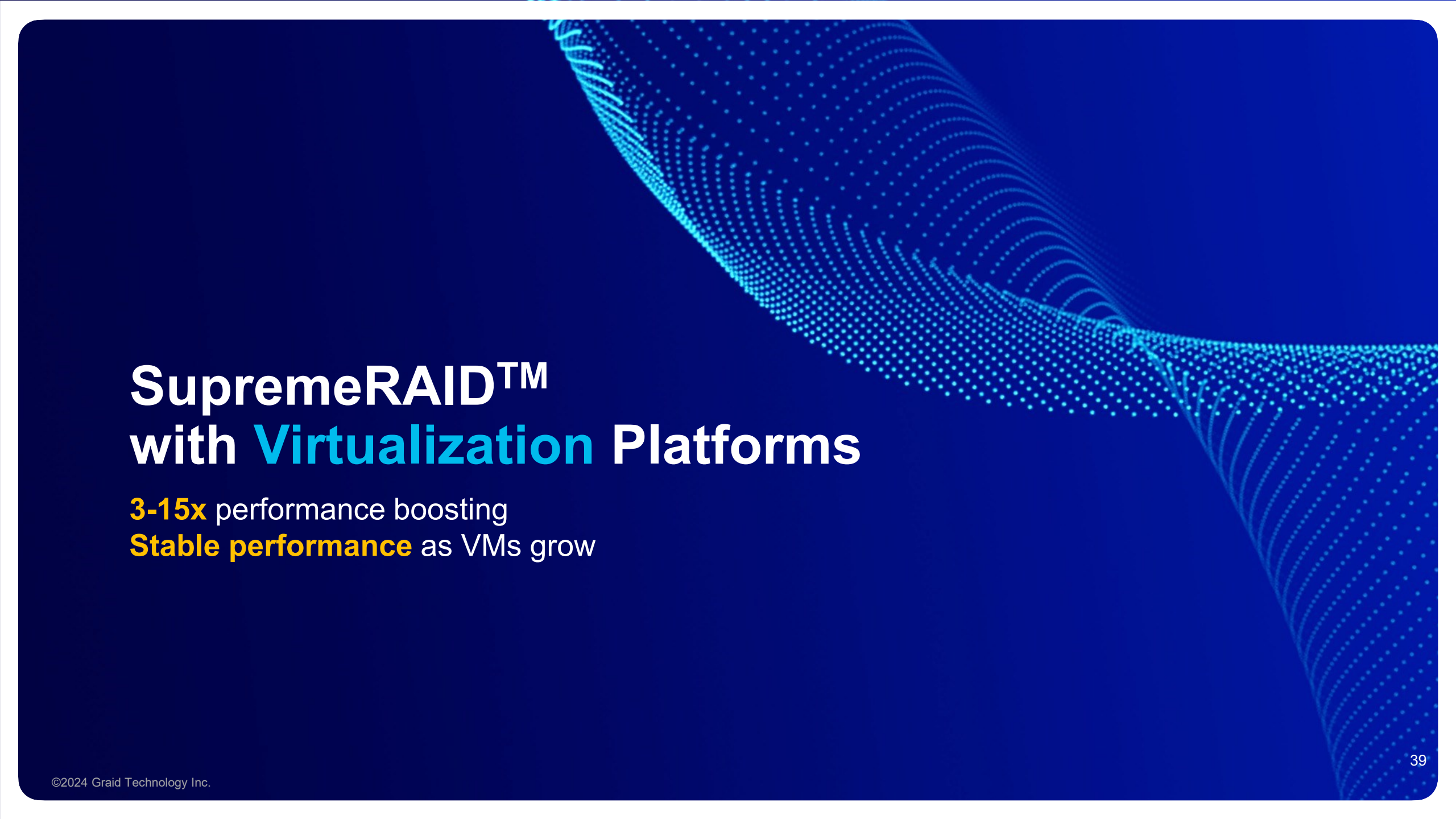
- Using **4 NVMe SSDs for RAID 5** and 15 15,000 RPM SAS HDDs for RAID 6, real-world performance **increased fourfold**
- For physical disks, most were in standby mode during the testing period.
- In standby mode, 4 NVMe SSDs compared to 15 SAS HDDs **saved 75% power consumption**.
- SupremeRAID™ offers the most cost-effective, highest-performance, and energy-efficient solution.

— G.M. of Tier I partner with Ansys



Potential market: Workstation





SupremeRAID™ with **Virtualization** Platforms

3-15x performance boosting

Stable performance as VMs grow

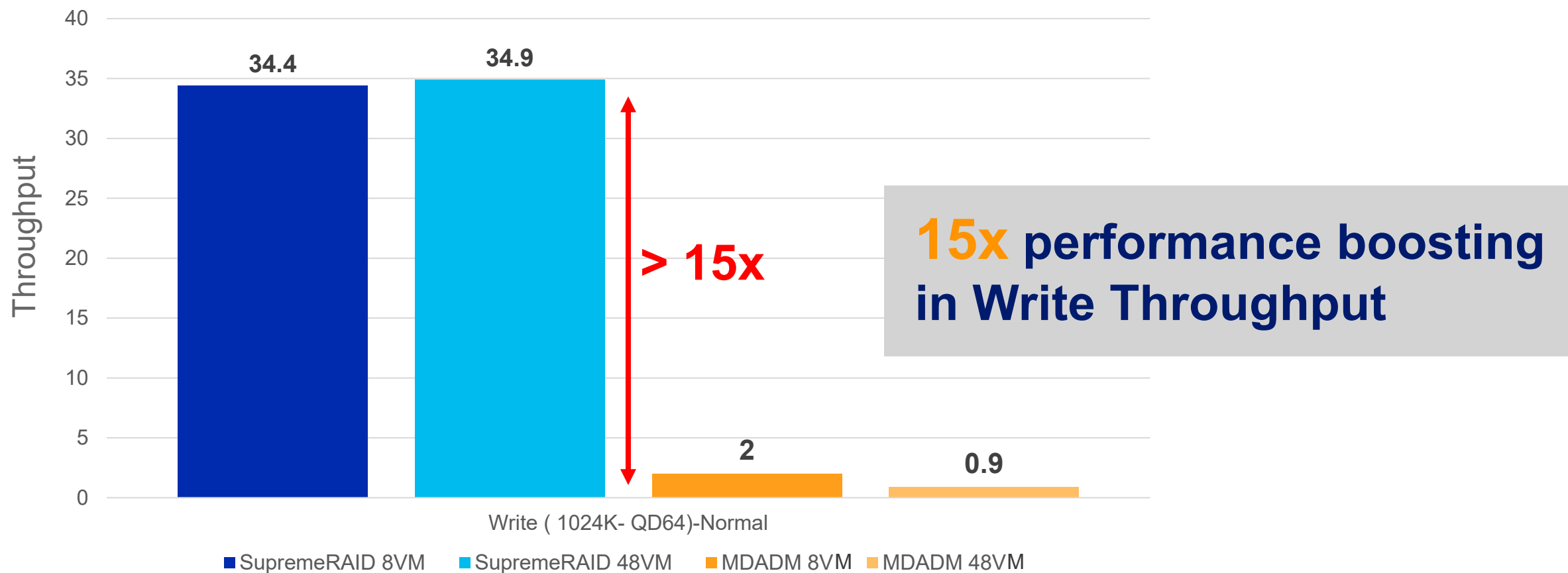
Platforms We Support



Write performance Comparison

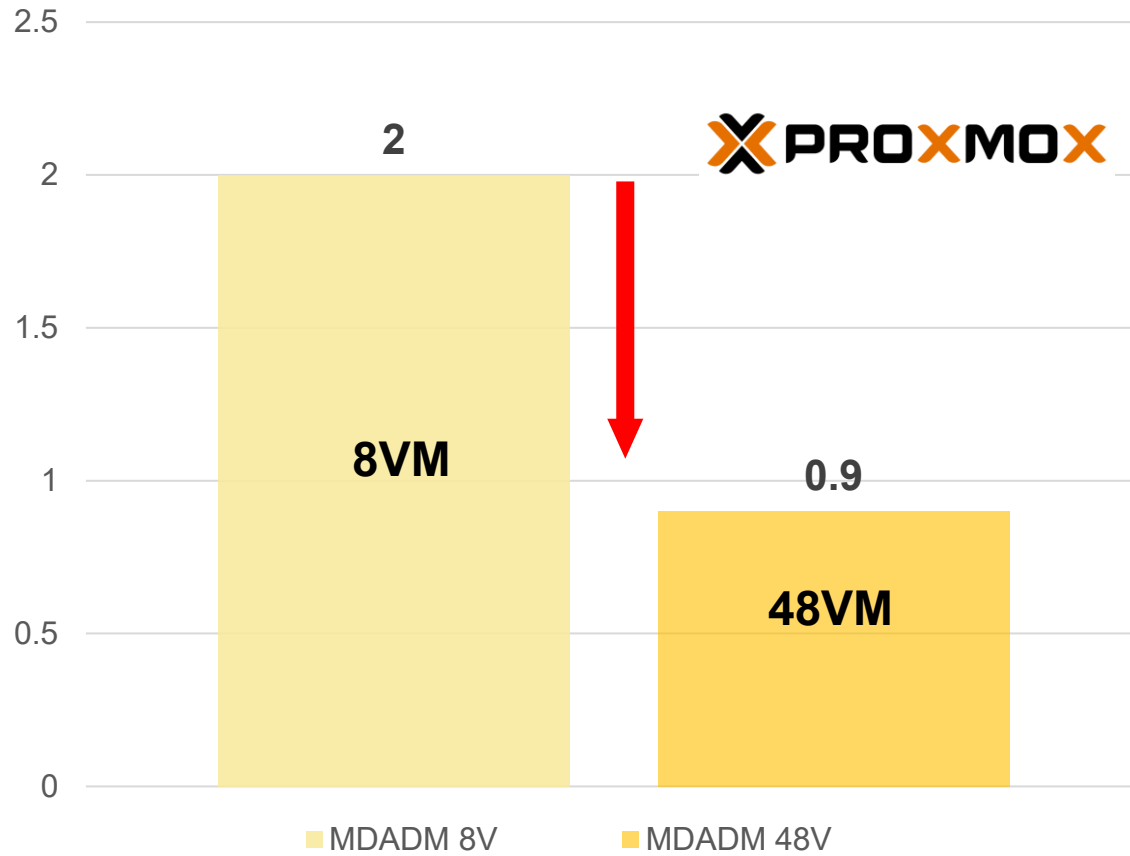


Aggregate FIO Performance Summary(Sequential IO)
8VM(6Disk) V.S. 48(1Disk)



MDADM Raid unstable when VM increased

Aggregate FIO Performance Summary(Sequential IO)
8VM(6Disk) V.S. 48(1Disk)

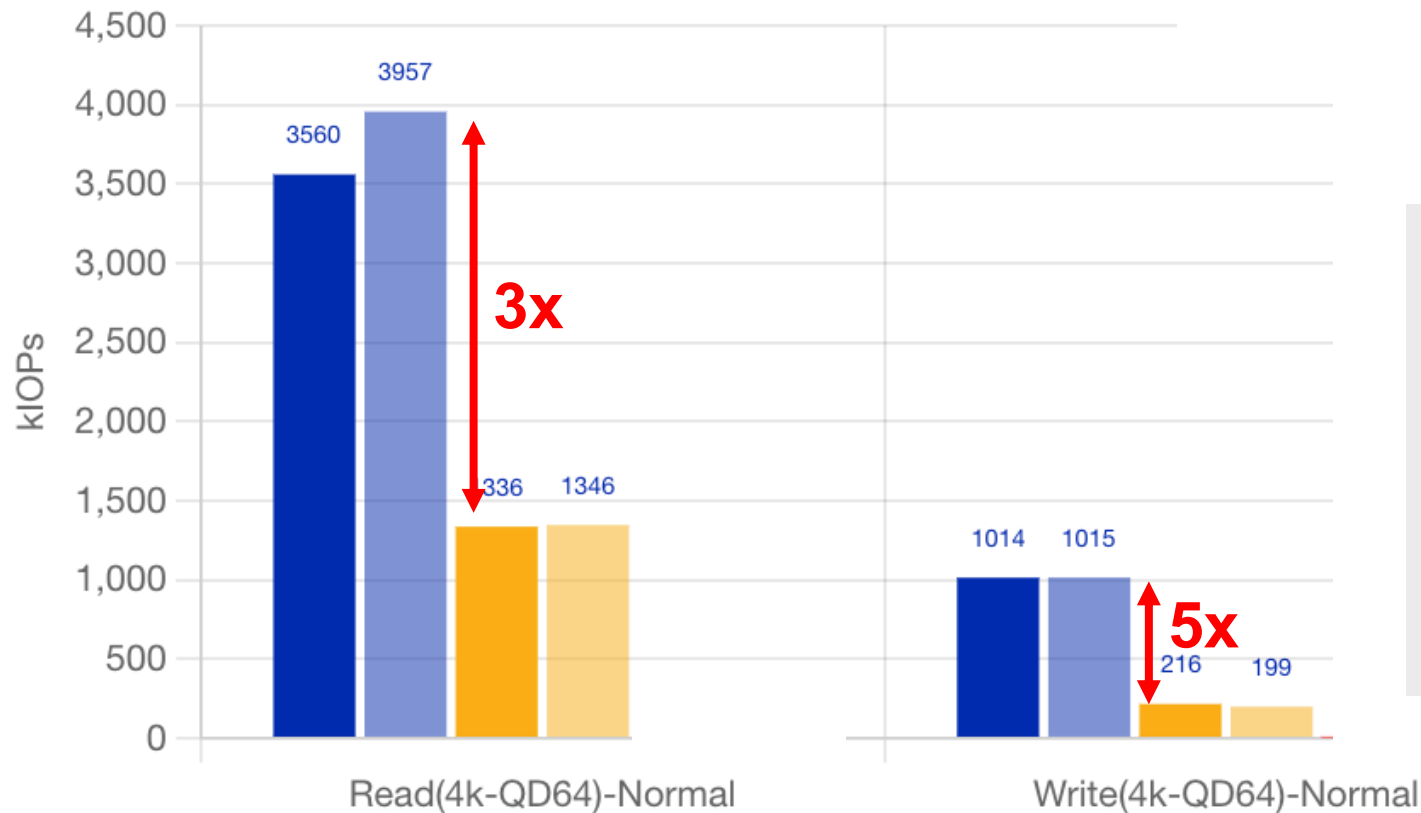


MDADM shows **unstable**
VM increase will consume more
CPU and MD raid compete with
CPU cycle.

VM performance Comparison



Aggregate FIO Performance Summary (Random IO) 8VM(6D) V.S. 48(1D)



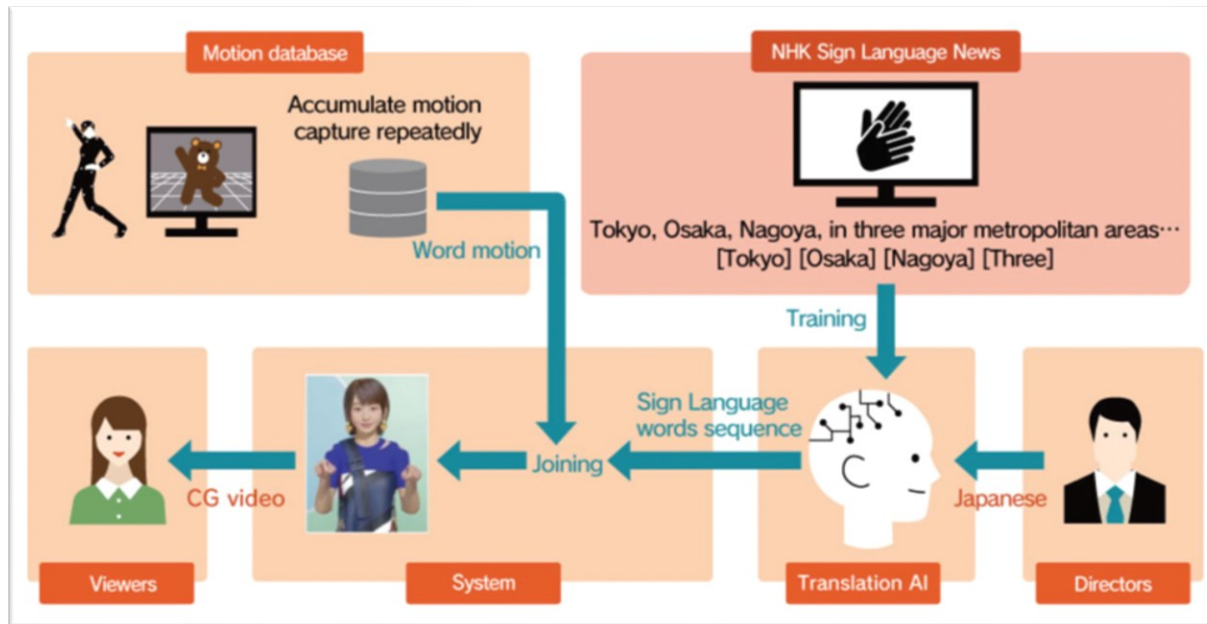
Significant Enhancement in Random IOPs

Random IOPs are enhanced by **3-5 times** over Linux MD, accelerating data access speed.

Media/ Content Broadcasting

Broadcasting Center CG creation

There is a bottleneck on random write performance when processing the CG creation, and Graid is in the POC progress now.



Animation Production in 8K/16K

The 16K program's audio and video content is **16x bigger** than the 4K content, offering a superior viewing experience but requiring better capability and efficiency in document preparation, processing, signal transmission, and scheduling.

SupremeRAID's 260GB / 100GB sequential R/W and 28M / 2M random R/W performance effectively enhance animation production efficiency.





Questions?
Thank You

PRESENTED BY:

[GRAIDTECH.COM](https://graidtech.com)